Research Assignment: Analysing Workflow and Process Design Using Activity Diagram

1 Question

1. Why is workflow/process analysis important before developing a software sys-tem?

Explain how process understanding affects software functionality and user experience.

Workflow or process analysis is important because software is built to support real-world activities. If

developers do not fully understand how users perform tasks, the software may automate the wrong

process, create confusion, or introduce inefficiencies.

Process analysis helps developers:

- understand how work currently happens,
- identify user requirements accurately,
- detect unnecessary or duplicated activities,
- reduce development mistakes,
- improve system efficiency and usability.

Without process analysis, developers may create technically correct software that still fails

operationally because it does not match how users actually work.

2. How can Activity Diagrams help identify problems in a system process

before implementation? Discuss issues such as:

- missing steps
- repeated activities
- incorrect decisions or inefficient workflows.

Understanding the process directly affects both functionality and user experience (UX).

Functionality

Process understanding ensures the system includes the correct features and business rules.

Examples:

- approval systems require authorization logic,
- ATM systems require balance verification before withdrawal,
- online shopping systems require inventory checking before payment.

If the process is misunderstood, important functions may be missing.

User Experience

Good workflow design reduces:

- unnecessary clicks,
- waiting time,
- confusion,
- repeated data entry.

A system aligned with user workflow feels intuitive.

3. Many software systems fail because of poor process flow rather than coding errors. Do you agree with this statement? Why or why not?

Provide a real example in this statement!

An Activity Diagram visually represents the flow of activities, decisions, and interactions inside a system process. Because the workflow becomes visible, problems can be detected before coding

begins.

a) Missing Steps

The diagram can reveal activities that were forgotten.

Example:

In an online shopping process:

- customer selects item,
- customer pays,
- order ships.

If there is no “inventory availability check,” the system could accept payments for unavailable products.

b) Repeated Activities

Activity Diagrams can reveal duplicated work.

Example:

A hospital registration system may ask patients to enter personal information multiple times across

departments. The diagram exposes redundancy that can be simplified.

c) Incorrect Decisions

The diagram can reveal decision logic errors.

Example:

In an ATM process:

- f balance is insufficient,
- system should reject withdrawal.

If the decision branch is missing or incorrectly connected, the system could allow overdrawing.

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d) Inefficient Workflow

Diagrams help identify unnecessary loops or bottlenecks.

Example:

A school registration system requiring manual approval from three departments sequentially may delay enrolment unnecessarily. Visualization helps redesign the flow for parallel approval.

4. Choose a real-world system (ATM, food delivery, online shopping, hospital, school registration, etc.) and analyse:

- **the main workflow/process**
- **possible bottlenecks or workflow problems, and how Activity Diagrams could improve the process efficiency.**

I largely agree with this statement, although coding errors still matter.

Many real-world software failures occur because:

- the workflow does not match user behaviour,
- business rules are misunderstood,
- processes are overly complex,
- communication between departments is poor.

A system can be technically stable but still fail operationally.

Real Example: Healthcare Systems

Some electronic medical record systems in hospitals created serious workflow problems:

- doctors needed excessive clicks,
- important patient data was difficult to access,
- workflows interrupted emergency treatment.

The software itself often functioned correctly from a programming perspective, but poor process design reduced efficiency and increased user frustration.

Another example is airport check-in systems:

Even if the software runs without crashes, poor workflow design can create long queues and passenger confusion.

So, software success depends not only on code quality but also on workflow quality.

5. How do decisions and branching processes affect system behaviour in Activity Diagrams?

Explain why decision points are important in workflow modelling with real case!

Customer Workflow

1. Browse products
2. Add item to cart
3. Proceed to checkout
4. Enter shipping information
5. Make payment
6. Receive confirmation
7. Seller processes order
8. Product shipped
9. Customer receives item

Possible Bottlenecks or Workflow Problems

a) Payment Delays

If payment verification is slow, users may abandon purchases.

b) Inventory Problems

If stock updates are delayed:

- customers may buy unavailable products,
- refunds become necessary.

c) Repeated Data Entry

Customers may repeatedly enter addresses or payment information.

d) Approval Delays

Manual order verification can slow shipping.

How Activity Diagrams Improve Efficiency

Activity Diagrams help by:

- visualizing the entire order flow,
- identifying unnecessary activities,
- simplifying customer interactions,
- improving decision logic.

Example improvements:

- automatic stock checking before payment,
- saved customer profiles,
- parallel payment and inventory validation,
- automated shipping notifications.

This reduces:

- processing time,
- human error,
- customer frustration.